



# Lakshya JEE 2.0 (2024)

## Limit, Continuity and Differentiability

**DPP-01**

1. If  $f(x) = \left( \frac{x^3 - 2x^2}{x^2 - 5x + 6} \right)$ , then  $\lim_{x \rightarrow 2} f(x)$  is:
- (1) 4 (2) -4  
(3) 2 (4) None of these

2.  $\lim_{x \rightarrow 1} \frac{x^{15} - 1}{x^{10} - 1}$  is equal to:

- (1)  $\frac{3}{2}$  (2)  $\frac{2}{3}$   
(3) 5 (4)  $-\frac{3}{2}$

3.  $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$  is equal to:

- (1)  $\frac{1}{2}$  (2) -2  
(3)  $-\frac{1}{2}$  (4) 4

4.  $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{\sqrt{2x}}$  is equal to:

- (1) 1 (2) 2  
(3) -1 (4) Does not exist

5.  $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x}$  is equal to:

- (1)  $\frac{1}{4}$  (2)  $\frac{3}{2}$   
(3) 1 (4) 2

6. Find the limit  $\lim_{x \rightarrow 0^+} \frac{\sqrt{x^3}}{\sqrt{x}}$

- (1) 0 (2) 1  
(3) 2 (4) None of these

7.  $\lim_{x \rightarrow 0} \frac{1 - \cos Ax}{x^2}$  is equal to:

- (1)  $A^2$  (2)  $\frac{A^2}{4}$   
(3)  $\frac{A^2}{2}$  (4) None of these

8. If  $\lim_{x \rightarrow 0} \frac{\sin 2x}{\tan\left(\frac{x}{2}\right)} = L_1$  and  $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} = L_2$  then the value of  $L_1 L_2$  is:

- (1) 4 (2) 8  
(3) 6 (4) 2

9.  $\lim_{x \rightarrow 0} \frac{2^x - 1}{x} + \lim_{x \rightarrow 0} \frac{3^x - 1}{x} - \lim_{x \rightarrow 0} \left( \frac{6^x - 1}{x} \right)$  is equal to:

- (1) 1 (2) 2  
(3) 0 (4) 4

10.  $\lim_{x \rightarrow 0} \frac{\sin 2x}{2 - \sqrt{4 - x}}$  is equal to:

- (1) 2 (2) 4  
(3) 8 (4) 0

11.  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x\sqrt{x^2}}$  is equals to:

- (1)  $\frac{1}{2}$  (2)  $-\frac{1}{2}$   
(3) 0 (4) Does not exist

12. Let  $\alpha$  and  $\beta$  be the distinct roots of  $ax^2 + bx + c = 0$ ,

then  $\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$  is equal to:

- (1)  $\frac{a^2}{2}(\alpha - \beta)^2$  (2) 0  
(3)  $-\frac{a^2}{2}(\alpha - \beta)^2$  (4)  $\frac{1}{2}(\alpha - \beta)^2$



13.  $\lim_{x \rightarrow 0} \left( \frac{x}{\tan^{-1} 2x} \right) =$

- (1) 0                      (2)  $\frac{1}{2}$   
(3) 1                      (4)  $\infty$

14.  $\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\sin x}$  is

- (1) 0                      (2) 1  
(3) 2                      (4) Non-existent

15.  $\lim_{x \rightarrow 0} \frac{\cos(\sin x) - 1}{x^2} =$

- (1) 1                      (2) -1  
(3) 1/2                      (4) -1/2

16.  $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$  is equal to :

- (1)  $-\pi$                       (2)  $\pi$   
(3)  $\frac{\pi}{2}$                       (4) 1

17.  $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x}$  is equal to

- (1) 2                      (2)  $\frac{1}{2}$   
(3) 4                      (4) 3

18.  $\lim_{x \rightarrow 0} \frac{\sin \alpha x}{\tan \beta x}$  is equal to:

- (1)  $\frac{2\alpha}{\beta}$                       (2)  $\frac{\beta}{\alpha}$   
(3)  $\frac{\alpha}{\beta}$                       (4) none of these

19.  $\lim_{x \rightarrow 0} \frac{\sin(\ln(1+x))}{\ln(1+\sin x)}$  is equal to:

- (1) 2                      (2) 1  
(3) -1                      (4)  $\frac{1}{2}$

20.  $\lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1} =$

- (1) 2                      (2)  $\ln 2$   
(3)  $\frac{\ln 2}{2}$                       (4)  $2 \ln 2$

**Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.**

## Answer Key

1. (2)
2. (1)
3. (1)
4. (4)
5. (4)
6. (1)
7. (3)
8. (2)
9. (3)
10. (3)

11. (4)
12. (1)
13. (2)
14. (3)
15. (4)
16. (2)
17. (1)
18. (3)
19. (2)
20. (4)



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